# Edugenie: Google Gemini Powered Learning Assistant

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## Topic: Functional Testing

## 1. Introduction

Functional testing is a software testing process used to verify whether the Edugenie application performs its required functions correctly. It checks the system from the user's point of view and ensures that each feature produces the expected result.

## 2. Objectives of Functional Testing

- Verify that all major Edugenie features work as expected.
- Check whether student questions are accepted correctly.
- Verify that requests are sent to the integrated AI service.
- Check whether useful responses are displayed correctly.
- Identify functional errors before the application is released.

## 3. Functional Test Cases

| Test Case | Input / Action | Expected Result |
|---|---|---|
| TC01 – Open Application | Launch Edugenie | Application opens successfully. |
| TC02 – Question Input | Enter a valid question | Question is accepted and displayed. |
| TC03 – Empty Input | Submit without a question | Suitable validation message is shown. |
| TC04 – AI Request | Submit a valid question | Request is processed successfully. |
| TC05 – AI Response | Wait for response | Generated response is displayed. |
| TC06 – Multiple Questions | Submit different questions | Each question receives an appropriate response. |
| TC07 – Network Error | Disconnect network and submit | Error/retry message is displayed. |
| TC08 – Clear/Reset | Use clear or reset option | Input/output is cleared correctly. |

## 4. Functional Testing Process

1. Identify the required functionality of each module.
2. Prepare test cases based on the expected behavior.
3. Provide valid and invalid inputs.
4. Execute the application and observe the output.
5. Compare the actual result with the expected result.
6. Record defects and retest after corrections.

## 5. Modules Tested

- User interface and navigation
- Question input and validation
- Google Gemini integration
- Response generation and display
- Error handling
- Clear/reset functionality

## 6. Expected Results

All major functions of Edugenie should operate according to the project requirements. Valid student questions should be processed correctly, AI-generated responses should be displayed, and invalid inputs or service failures should produce suitable error messages.

## 7. Benefits of Functional Testing

- Improves application reliability.
- Detects functional defects early.
- Ensures requirements are satisfied.
- Improves the student user experience.
- Helps verify the complete learning workflow.

## 8. Conclusion

Functional testing is an important stage in the Edugenie project. It confirms that the learning assistant, user interface, AI integration, response handling, and error handling work correctly. Successful functional testing increases confidence that Edugenie is ready for practical use.